

Wave Motion

Pitch: It is the perception of sound frequency by human ears.

Higher Frequency Sound: High pitch

Low Frequency Sound: High pitch



(a) Audible frequency: $f: 20 \text{ Hz} - 20 \text{ KHz}$

(b) Infrasonic frequency: $f < 20 \text{ Hz}$

(c) Ultrasonic frequency $f > 20 \text{ KHz}$

Loudness of sound waves (L)

It is the perception of human mind towards its intensity.

Let us assume I_0 is the lower limit of human hearing.

$$I_0 = 10^{-12} \text{ W / m}^2$$

The expression of loudness (L) can be written as

$$L(\text{dB}) = 10 \log_{10} \left(\frac{I_1}{I_0} \right)$$

The relative loudness can be written as

$$L(\text{dB}) = 10 \log_{10} \left(\frac{I_1}{I_2} \right)$$

Standing waves:

$$y_1 = A \sin(\omega t - kx), \quad y_2 = -A \sin(\omega t + kx)$$

The resultant wave can be written as

$$y = y_1 + y_2 = A \sin(\omega t - kx) - A \sin(\omega t + kx)$$

$$y = y_1 + y_2 = -2A \sin(kx) \cos(\omega t)$$

$$y = -2A \sin\left(\frac{2\pi}{\lambda} x\right) \cos\left(\frac{2\pi}{\lambda} vt\right)$$

The above equation represents the standing wave. Net energy transfer is zero.

$$y_1 = A \sin(\omega t - kx), \quad y_2 = A \sin(\omega t + kx)$$

$$y = y_1 + y_2 = 2A \cos\left(\frac{2\pi}{\lambda} x\right) \sin\left(\frac{2\pi}{\lambda} vt\right)$$

Energy of stationary wave

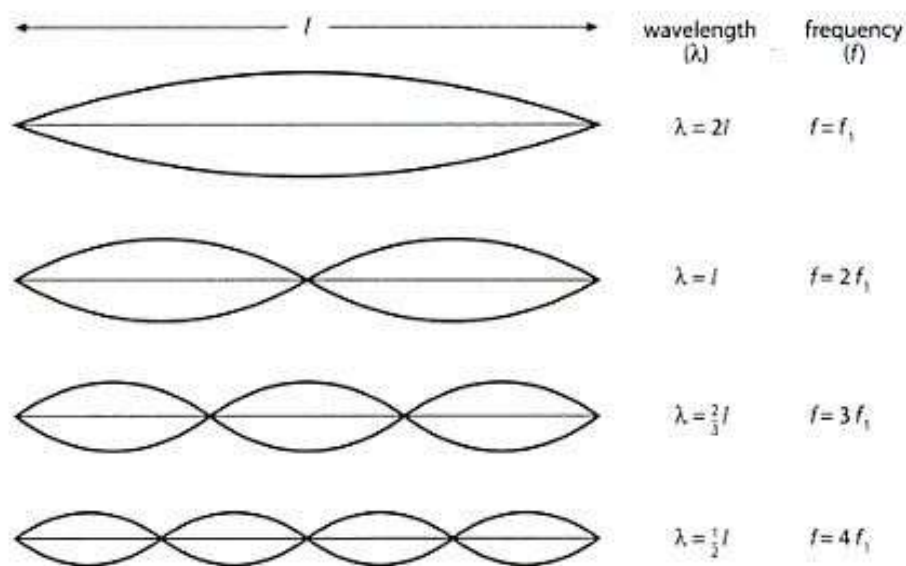
$$E = \mu \omega^2 A^2 L; \quad \mu = \text{Mass} / \text{Length}$$

$$\text{Energy per unit length: } \frac{E}{L} = \mu \omega^2 A^2$$

Example: 1 Clamped string

$$\text{Fundamental frequency / First harmonics: } \frac{\lambda_0}{2} = L \Rightarrow \lambda_0 = 2L$$

$$f_0 = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$



$$\text{First overtone: } f_1 = \frac{2}{2L} \sqrt{\frac{T}{\mu}} = 2f_0$$

$$\text{Second overtone: } f_3 = \frac{3}{2L} \sqrt{\frac{T}{\mu}} = 3f_0$$

$$\text{General formula: } f_p = \frac{p}{2L} \sqrt{\frac{T}{\mu}}$$

Example: 2 Clamped at end

$$\text{Fundamental frequency: } \frac{\lambda_0}{4} = L \Rightarrow \lambda_0 = 4L$$

$$f_0 = \frac{1}{4L} \sqrt{\frac{T}{\mu}}$$

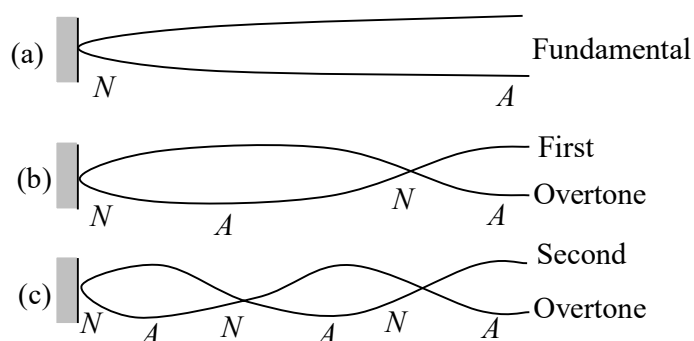
First Overtone: $\frac{3\lambda_1}{4} = L \Rightarrow \lambda_1 = \frac{4}{3}L$

$$f_1 = \frac{3}{4L} \sqrt{\frac{T}{\mu}} = 3f_0$$

Second Overtone

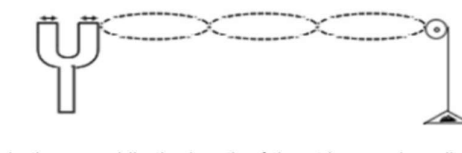
$$\frac{5\lambda_2}{4} = L \Rightarrow \lambda_2 = \frac{4}{5}L$$

$$f_2 = \frac{5}{4L} \sqrt{\frac{T}{\mu}} = 5f_0$$



Melde's experiment

Melde's experimental set up consists of a light string tied to one of the prongs of a tuning fork which is mounting on a sounding board. The other end of the string is passed over a horizontal pulley and a light pan is suspended from the adjusted by changing the weights placed in the pan while the length of the string can be adjusted by changing the position of the pulley towards or away from the fork. If the fork is adjusted so that its arms vibrate parallel to the length of the string. Through adjusting the length or the tension number of clear loops can be visible as the standing wave is formed. When the prongs vibrate parallel to the length of string, the frequency of the fork is twice the frequency of the string. Since the string is vibrating in the fundamental mode, its frequency " n " is given by



$$n = \frac{1}{2L} \sqrt{\frac{T}{\mu}}, \text{ Therefore, the frequency of the tuning fork will be } N = 2n = \frac{1}{L} \sqrt{\frac{T}{\mu}}$$

If number of loops formed is " p ", then $N = \frac{p}{L} \sqrt{\frac{T}{\mu}}$.